Introduction To Programming

Python fundamentals

Universidad...

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Today's Goals

- Fundamentals
- Type of Variables
- Your first program

Fundamentals

Computers are obedient, not smart. They do exactly what you tell them to do, word for word, step by stepeven if your instructions lead to a error.
An **algorithm** is a precise, step-by-step set of instructions designed to solve a specific problem or complete a task.

Fundamentals

Computers Do What You Tell Them.

Think a computer of a kitchen assistant.If you tell it to "cook a complex dish". It will freeze because it doesn't know what that means. But if you give it a step-by-step recipe like: "chop 5 tomatoes, blend for 10 seconds, add 2 grams of salt", it will do it very quickly; because it moves at millions of operations per second.

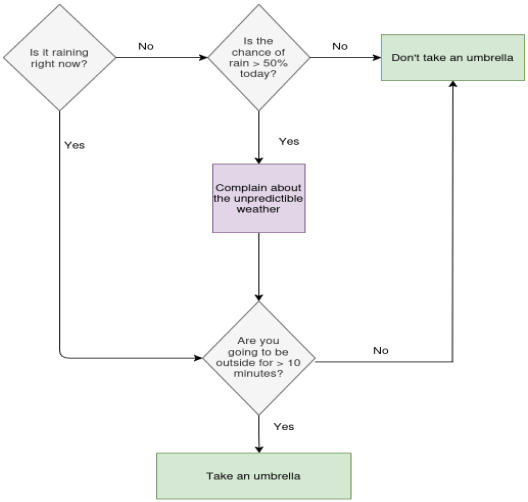
You Should Be Familiar With...

The Concept Of:

- Variable
- True/false statements
- Connecting statements (or,and, not)
- Making decisions (if ... then)
- Following flowcharts

If you shouldn't be familiar with the Flowcharts

You should be familiar with following a set of instructions and making branching desicions.



The structured program theorem

States that any computable algorithm can be expressed only with three logic control structures.

- Sequence or simple action: Executing one instruction after another in a specific, linear order.
- Conditional: valuating a boolean condition to "choose the correct path".
- Loop or iteration: Repeating a block of instructions as long as a certain condition remains true.

Variables

The = operator executes a **right-to-left assignment** (evaluating the right side first, then binding to the left name), not an algebraic equivalence.

Pitfalls & Invalid Code

Math equivalence: Treating = as static math logic (`a + x = b` throws `SyntaxError`).

Syntax violations: Starting with numbers (`1st = 5`), spaces, or overriding keywords (`print = 10`).

Single-letter cryptics: Using `x`, `t`, `p` without domain context.

Correct Model & Standards

Right-to-left binding: `x = x + 1` evaluates the new value first, then moves the name tag.

Valid naming: Letters or underscores (_), case-sensitive (`age` $\neq$ `Age`).

Style: Descriptive `snake_case` names (`circle_area`), $<$ 80 chars/line.

Exercise

"Allowed, Incorrect, Good Variables Names"

1.- customer_list
2.- print (where this word is a color other than black in Spyder)
3.- list_of_things_I_need_to_pick_up_from_the_store
4.- %score
5.- 1
6.- A